

McGill University
ECSE 222 – Digital Logic (Fall 2025)
Pen-and-Paper Assignment #5

Instructions

- TA in-charge: Abidur Rahman (abidur.rahman@mail.mcgill.ca)
- Write/type your answers in the space provided.
- Due date: Please refer to myCourses.
- You must electronically submit the assignment in the form of a PDF file on myCourses before the deadline. You may submit as many times as you want, however, only the last submission will be kept by the system.
- You must solve the assignment in teams of two students, whereas only one of the students should perform the actual electronic submission.
- Late submissions will incur penalties as described in the course syllabus.

Student identification

First student full name:	Aneurin Hullah
First student ID number:	261283866
Second student full name:	Alexandre Pujal
Second student ID number:	261183554
Third student full name (if applicable)	
Third student ID number (if applicable)	

Evaluation (for course graders use only)

Question 1 (20%)	Question 2 (20%)	Question 3 (20%)	Total
Question 4 (20%)	Question 5 (20%)	-	
		-	

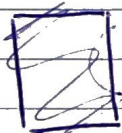
ECSE 222 - Pen and Paper Assignment #5

Alexandre PUJAL

261183554

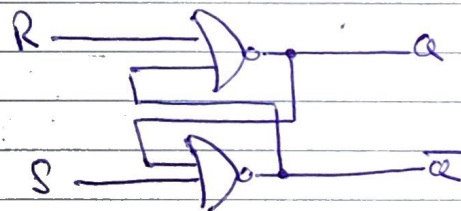
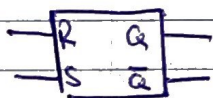
Question 1:

~~Construct the truth table for the SR latch.~~



$Q(t)$	S	R	$Q(t+1)$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

SR latch:



Characteristic Table:

S	R	$Q(t+1)$
0	0	$Q(t)$
0	1	0
1	0	1
1	1	$\emptyset$

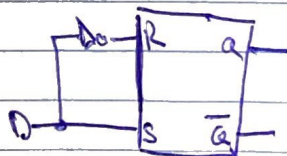
$$\left. \begin{array}{l} 0 \ 0 \ Q(t) \\ 0 \ 1 \ 0 \end{array} \right\} = \bar{R}Q$$

$Q(t+1)$

$$= \bar{S}\bar{R}Q + SS$$

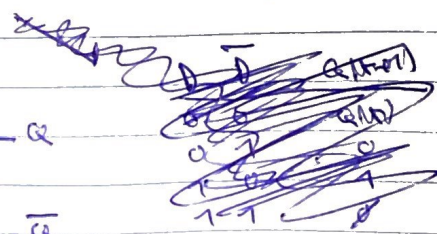
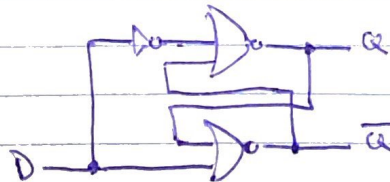
$$= \bar{R}Q + S$$

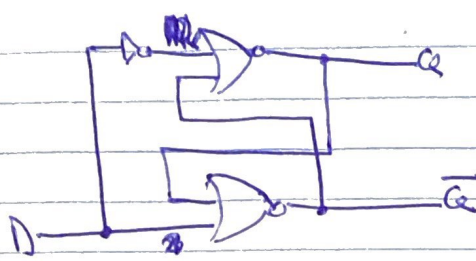
D latch:



D	$Q(t+1)$
0	0
1	1

If we want the D-latch to behave like an SR latch, we need $D = S + \bar{R}Q$

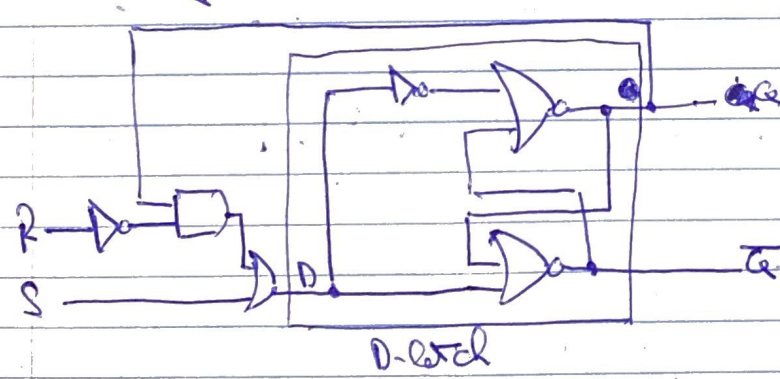
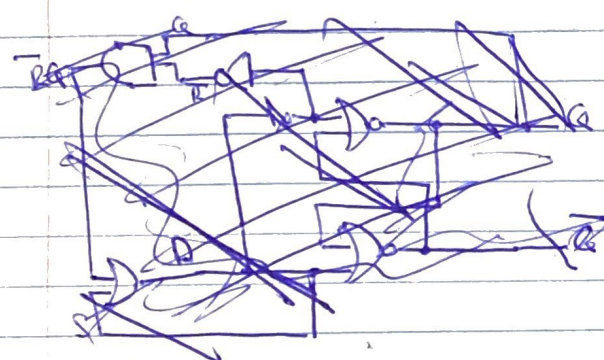




$$Q = S + \overline{R}Q$$

setting D to be $S + \overline{R}Q$

~~The D-latch is originally designed so that S and R are never equal as S is originally R-bar and R is originally S~~



SR-latch implemented using a D latch, an additional NOT, AND, and OR gates

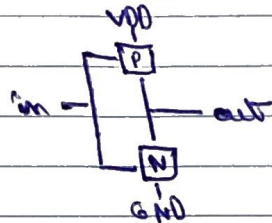
Question 2:

Truth table P switch:

C	Switch status
0	ON state (closed)
1	OFF state (open)

Truth table N switch:

C	Switch status
0	OFF state (open)
1	ON state (closed)



If $in = '0'$ then N switch is OFF and P switch is ON
→ $out = '1'$

If $in = '1'$ then N switch is ON and P switch is OFF
→ $out = '0'$

	x_1	x_2	x_3	F
0	0	0	0	1
1	0	0	1	1
2	0	1	0	1
3	0	1	1	1
4	1	0	0	1
5	1	0	1	0
6	1	1	0	1
7	1	1	1	0

Hence, $F = (\overline{x_1} + \overline{x_2} + \overline{x_3})(\overline{x_1} + \overline{x_2} + x_3)(x_1 + x_2 + \overline{x_3})$

x_3	$x_1 x_2$	00	01	11	10
0	0	1	1	0	1
1	1	1	1	0	0

$F = (\overline{x_1} + \overline{x_2})(\overline{x_1} + \overline{x_2} + \overline{x_3})$

$$= \overline{x_1} \overline{x_2} \overline{x_3} + \overline{x_1} \overline{x_2} \overline{x_3} + \overline{x_1} \overline{x_2} \overline{x_3} + \overline{x_1} \overline{x_2} \overline{x_3}$$

$$\text{absorption } \frac{\overline{x_1} \overline{x_2} + \overline{x_1} \overline{x_2} \overline{x_3} + \overline{x_1} \overline{x_2} \overline{x_3} + \overline{x_1} \overline{x_2} \overline{x_3}}{\overline{x_1} \overline{x_2} + \overline{x_1} \overline{x_2} \overline{x_3} + \overline{x_1} \overline{x_2} \overline{x_3}}$$

$$\frac{\overline{x_1} \overline{x_2} + \overline{x_1} (\overline{x_3} + x_3) + \overline{x_2} \overline{x_3}}{\overline{x_1} \overline{x_2} + \overline{x_1} \overline{x_2} \overline{x_3} + \overline{x_1} \overline{x_2} \overline{x_3}}$$

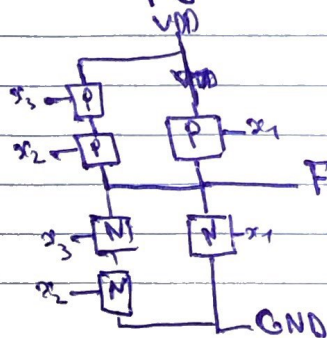
$$= \overline{x_1} + \overline{x_2} \overline{x_3}$$

$$= (\overline{x_1} + \overline{x_2})(\overline{x_1} + \overline{x_3})$$

minimal POS form

An OR gate would imply two parallel branches

An AND would imply series switches



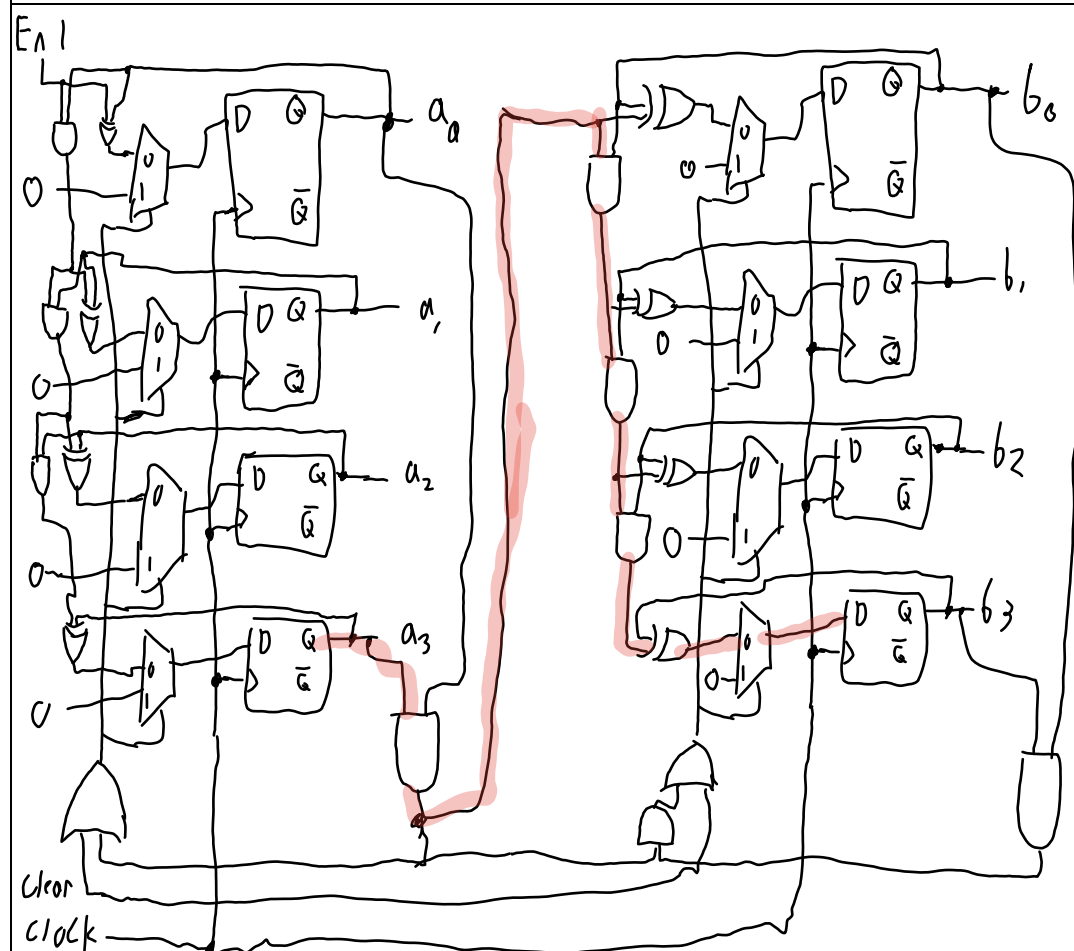
Hence, the implementation is the following

Question 3 (20%)

Determine the maximum operating frequency of a two-digit BCD up counter (as shown in class). Provide the circuit diagram of the counter and mark the critical path of the circuit.

Given the following specifications:

- FFs setup time is 3 ns
- FFs hold time is 1 ns
- FFs t_{CQ} and propagation delay of all gates and logic blocks is 3 ns
- The maximum fan-in of all gates is 2

Answer

Critical Path, a red

$$T \geq t_{CQ} + t + t_{su}$$

$$T \geq 3ns + (3ns \cdot 6 \text{ gates/blocks}) + 3ns = 24ns$$

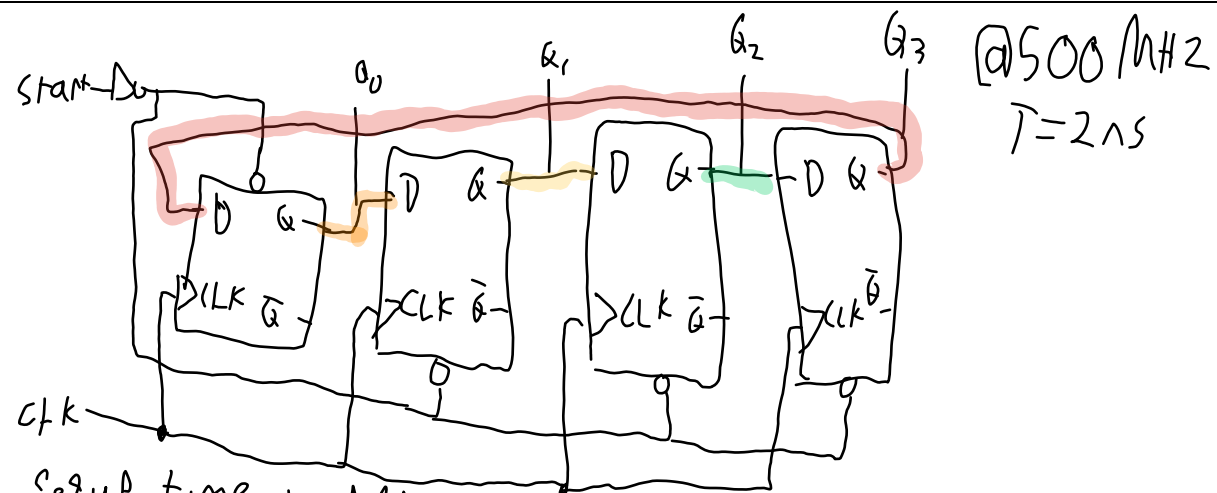
$$\text{Frequency max is } 41.6 \text{ MHz} = \frac{1}{24ns}$$

Question 4 (20%)

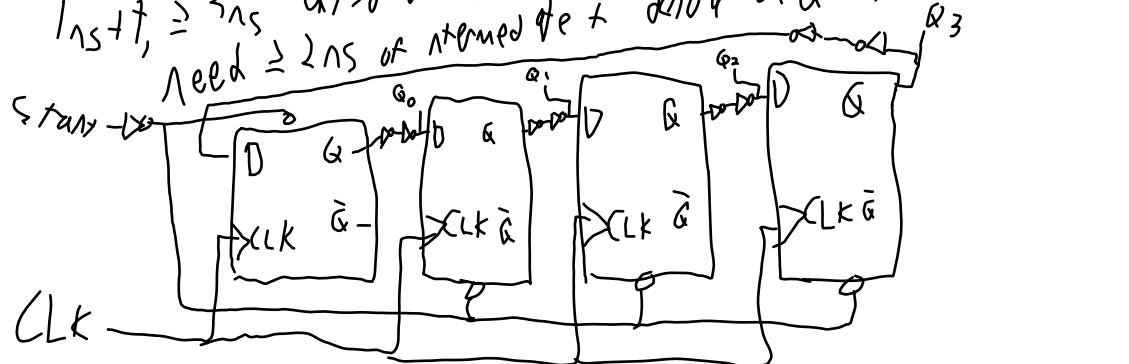
The 4-bit ring counter shown in class (implemented using DFFs) is operated using a clock frequency of 500 MHz, check the circuit for timing violations (show your work). If violations exist, correct the circuit using as few logic gates as possible (provide new schematic). Prove that all timing violations are solved and find the new operating frequency.

Given the following timing constraints:

- FFs setup time is 4 ns
- FFs hold time is 3 ns
- FFs t_{CQ} is 1 ns
- Latency of any gate is equal to the fan-in of the gate in units of ns

Answer

Setup time violation here
 $t \geq t_{CQ} + t_i + t_{su}$ Violated setup time condition
 $2ns \geq 1ns + 0 + 4ns$ $T_{old} \geq 5ns$ $T_{new} \geq 1ns + 2ns + 4ns$
 hold time check
 $t_{CQ} + t_i \geq t_h$ $1ns + 0 \geq 3ns$ also violated
 $need \geq 2ns$ or inserted the delay on all lines highlighted



Question 5 (20%)

Design a 4-bit counter that counts in Excess-3 code. The counter should count to $(6)_{10}$ and then restart. The counter should have a synchronous (not an asynchronous) active-high reset signal that resets the counter to '0'. You may use any logic gates/blocks learned in class and the constants '0' and '1'.

Answer